

Robotix - Inventory Turnover Analysis

Q4 2024 Report

Date: January 15, 2025

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Distribution: Executive Team, Board of Directors

Executive Summary

Our Q4 2024 inventory turnover analysis shows a ratio of 3.65x, slightly above industry average but below our target of 4.0x. Total inventory value stands at \$7.2M, with Days Inventory Outstanding (DIO) at 100 days. While meeting current operational needs, opportunities exist to optimize working capital efficiency.

Key Metrics:

Overall Turnover Ratio: 3.65x
Days Inventory Outstanding: 100.0 days
Total Inventory Value: \$7.2M
Working Capital Impact: \$2.1M improvement opportunity

Inventory Turnover by Category

Product Category	Turnover Ratio	Avg. Inventory Value	DOI (Days)
Industrial Robots	4.2x	\$2.8M	86.9
Collaborative Robots	3.8x	\$1.9M	96.1
Mobile Robots	3.1x	\$1.4M	117.7
Components & Accessories	3.5x	\$1.1M	104.3

Detailed Analysis

Raw Materials Inventory

- Total Value: \$2.4M
- Turnover: 4.1x
- Key Components:
 - Motors & Actuators: \$820K
 - Controllers: \$640K
 - Sensors: \$520K

- Frames & Structural: \$420K

Work-in-Progress (WIP)

- Total Value: \$1.8M
- Average Assembly Time: 12 days
- Aging Profile:
 - 0-30 days: 65%
 - 31-60 days: 25%
 - 60 days: 10%

Finished Goods

- Total Value: \$3.0M
- By Product Line:
 - Industrial Series: \$1.4M
 - Collaborative Series: \$900K
 - Mobile Series: \$700K

Inventory Metrics

Metric	Q4 2024 Value
Beginning Inventory	\$7.8M
Purchases/Production	\$9.2M
COGS	\$9.8M
Ending Inventory	\$7.2M
Average Inventory	\$7.5M

Location Analysis

Minneapolis Manufacturing Facility

- Inventory Value: \$4.8M
- Turnover: 3.8x
- Capacity Utilization: 82%

Portland Distribution Center

- Inventory Value: \$2.4M
- Turnover: 3.3x
- Fill Rate: 94%

Quarter-over-Quarter Comparison

Quarter	Turnover Ratio	Total Inventory	DIO
Q4 2024	3.65x	\$7.2M	100.0
Q3 2024	3.52x	\$7.8M	103.7
Q2 2024	3.48x	\$7.5M	104.9
Q1 2024	3.41x	\$7.9M	107.0

Risk Factors

1. **Technology Obsolescence** - New controller platform launch in Q2 2025 - Legacy component phase-out risk: \$340K exposure
2. **Supply Chain** - Semiconductor lead times: 16-20 weeks - Single-source components: 12% of inventory value

Recommendations

1. **Inventory Optimization** - Reduce slow-moving inventory by \$800K through targeted promotions - Implement VMI program for top 3 suppliers - Target turnover ratio: 4.0x by Q2 2025
2. **Process Improvements** - Enhanced demand forecasting system implementation - Automated reorder point adjustments - Cross-dock operations for fast-moving items

Action Items

Priority	Action	Owner	Timeline
High	Slow-moving inventory reduction	Operations Dir.	Q1 2025
High	VMI program implementation	Procurement	Q2 2025
Medium	Forecast system upgrade	IT/Operations	Q3 2025

Cash Flow Impact

Working capital improvement potential: - Inventory reduction target: \$800K - Cash conversion cycle improvement: 8 days - Projected annual carrying cost savings: \$120K

This analysis identifies significant opportunities to optimize inventory management while maintaining service levels. Implementation of recommendations could release \$2.1M in working capital over the next two quarters.

Respectfully submitted,

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Robotix

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